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Semiconductor device structure and method for forming the same

Abstract

A method for forming a semiconductor device structure includes forming nanostructures over a front side of a substrate. The method also includes forming a gate structure surrounding the nanostructures. The method also includes forming a source/drain structure beside the gate structure. The method also includes forming a trench though the substrate from a back side of the substrate. The method also includes forming a first silicide layer in contact with the source/drain structure. The method also includes forming a second silicide layer over the first silicide layer and the sidewalls of the trench. The method also includes depositing a first conductive material over the second silicide layer. The method also includes etching back the first conductive material. The method also includes etching back the second silicide layer. The method also includes depositing a second conductive material in the trench.

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Background/Summary

BACKGROUND

(1) Semiconductor devices are used in a variety of electronic applications, such as personal computers, cell phones, digital cameras, and other electronic equipment. Semiconductor devices are typically fabricated by sequentially depositing insulating or ILD structures, conductive layers, and semiconductive layers of material over a semiconductor substrate, and patterning the various material layers using lithography to form circuit components and elements thereon. Many integrated circuits are typically manufactured on a single semiconductor wafer, and individual dies on the wafer are singulated by sawing between the integrated circuits along a scribe line. The individual dies are typically packaged separately, in multi-chip modules, for example, or in other types of packaging.

(2) Recently, multi-gate devices have been introduced in an effort to improve gate control by increasing gate-channel coupling, reduce OFF-state current, and reduce short-channel effects (SCEs). One such multi-gate device that has been introduced is the gate-all around transistor (GAA). The GAA device gets its name from the gate structure which can extend around the channel region providing access to the channel on two or four sides. GAA devices are compatible with conventional complementary metal-oxide-semiconductor (CMOS) processes.

(3) However, integration of fabrication of the GAA features around the nanowire can be challenging. While the current methods have been satisfactory in many respects, continued improvements are still needed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It should be noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIGS. **1**A-**1**N are perspective representations of various stages of forming a semiconductor device structure, in accordance with some embodiments of the disclosure.

(3) FIGS. **1**O-**1**W are cross-sectional representations of various stages of forming a semiconductor device structure, in accordance with some embodiments of the disclosure.

(4) FIG. **2** is an enlarged cross-sectional representation of a semiconductor device structure, in accordance with some embodiments of the disclosure.

(5) FIGS. **3**A-**3**B are cross-sectional representations of various stages of forming a semiconductor device structure, in accordance with some embodiments of the disclosure.

(6) FIGS. **4**A-**4**C are cross-sectional representations of various stages of forming a semiconductor device structure, in accordance with some embodiments of the disclosure.

(7) FIG. **5** is an enlarged cross-sectional representation of a semiconductor device structure, in accordance with some embodiments of the disclosure.

(8) FIG. **6** is an enlarged cross-sectional representation of a semiconductor device structure, in accordance with some embodiments of the disclosure.

DETAILED DESCRIPTION

(9) The following disclosure provides many different embodiments, or examples, for implementing different features of the subject matter provided. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or

on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.
(10) Some variations of the embodiments are described. Throughout the various views and illustrative embodiments, like reference numbers are used to designate like elements. It should be understood that additional operations can be provided before, during, and after the method, and some of the operations described can be replaced or eliminated for other embodiments of the method.
(11) The nanostructure transistor (e.g. nanosheet transistor, nanowire transistor, multi-bridge channel, nano-ribbon FET, gate all around (GAA) transistor structures) described below may be patterned by any suitable method. For example, the structures may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, smaller pitches than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the GAA structure.
(12) Embodiments for forming a semiconductor device structure are provided. The method for forming the semiconductor device structure may include forming a back-side source/drain contact structure by a two-step etching back process. The back-side source/drain contact structure may be formed by a bottom-up deposition process. The resistance may be minimized. In addition, the defect such as selective loss or void in the back-side source/drain contact structure may be prevented.
(13) FIGS. **1**A-**1**N are perspective representations of various stages of forming a semiconductor device structure **10***a*, in accordance with some embodiments of the disclosure. The semiconductor device structure **10***a* may be a gate all around (GAA) transistor structure. FIGS. **1**O-**1**W are cross-sectional representations of various stages of forming a semiconductor device structure **10***a*, in accordance with some embodiments of the disclosure.
(14) A semiconductor stack including first semiconductor material layers **106** and second semiconductor material layers **108** are formed over a substrate **102**, as shown in FIG. **1**A in accordance with some embodiments. The substrate **102** may be a semiconductor wafer such as a silicon wafer. The substrate **102** may also include other elementary semiconductor materials, compound semiconductor materials, and/or alloy semiconductor materials. Examples of the elementary semiconductor materials may include, but are not limited to, crystal silicon, polycrystalline silicon, amorphous silicon, germanium, and/or diamond. Examples of the compound semiconductor materials may include, but are not limited to, silicon carbide, gallium nitride, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide. Examples of the alloy semiconductor materials may include, but are not limited to, SiGe, GaAsP, AlinAs, AlGaAs, GalnAs, GaInP, and/or GaInAsP. The substrate **102** may include an epitaxial layer. For example, the substrate **102** may be an epitaxial layer overlying a bulk semiconductor. In addition, the substrate **102** may also be semiconductor on insulator (SOI). The SOI substrate may be fabricated by a wafer bonding process, a silicon film transfer process, a separation by implantation of oxygen (SIMOX) process, other applicable methods, or a combination thereof. The substrate **102** may be an N-type substrate. The substrate **102** may be a P-

type substrate.

(15) Next, first semiconductor material layers **106** and second semiconductor material layers **108** are alternating stacked over the substrate **102** to form the semiconductor stack, as shown in FIG. **1**A in accordance with some embodiments. The first semiconductor material layers **106** and the second semiconductor material layers **108** may include Si, Ge, SiGe, GaAs, InSb, GaP, GaSb, InAlAs, InGaAs, GaSbP, GaAsSb, or InP. The first semiconductor material layers **106** and second semiconductor material layers **108** may be made of different materials with different etching rates. In some embodiments, for example, the first semiconductor material layers **106** are made of SiGe and the second semiconductor material layers **108** are made of Si.

(16) The first semiconductor material layers **106** and second semiconductor material layers **108** may be formed by low pressure chemical vapor deposition (LPCVD) process, epitaxial growth process, other applicable methods, or a combination thereof. The epitaxial growth process may include molecular beam epitaxy (MBE), metal organic chemical vapor deposition (MOCVD), or vapor phase epitaxy (VPE).

(17) It should be noted that, although there are three layers of the first semiconductor material layers **106** and three layers of the second semiconductor material layers **108** shown in FIG. **1**A, the number of the first semiconductor material layers **106** and second semiconductor material layers **108** are not limited herein, depending on the demand of performance and process. For example, the semiconductor structure may include two to six layers of the first semiconductor material layers **106** and two to six layers of the second semiconductor material layers **108**.

(18) Next, a mask structure may be formed over the first semiconductor material layers **106**, as shown in FIG. **1**A in accordance with some embodiments. The mask structure may be made of silicon nitride, silicon carbon nitride (SiCN), or applicable material. The mask structure may be formed by a deposition process, such as low-pressure CVD (LPCVD) process, plasma enhanced CVD (PECVD) process, or another deposition process.

(19) After the first semiconductor material layers **106** and the second semiconductor material layers **108** are formed as the semiconductor material stack over the substrate **102**, the semiconductor material stack is patterned to form fin structures **104** using the mask structure as a mask layer, as shown in FIG. **1**B in accordance with some embodiments. The fin structures **104** may include the base fin structures **105** and the semiconductor material stacks, including the first semiconductor material layers **106** and the second semiconductor material layers **108**, formed over the base fin structures **105**.

(20) The patterning process may including forming a mask structure over the first semiconductor material layers **106** and the second semiconductor material layers **108** and etching the semiconductor material stack and the underlying substrate **102** through the mask structure, as shown in FIG. **1**B in accordance with some embodiments. The mask structure may be a multilayer structure including a pad layer and a hard mask layer formed over the pad layer. The pad layer may be made of silicon oxide, which may be formed by thermal oxidation or CVD. The hard mask layer may be made of silicon nitride, which may be formed by CVD, such as LPCVD or plasma-enhanced CVD (PECVD).

(21) The patterning process of forming the fin structures **104** may include a photolithography process and an etching process. The photolithography process may include photoresist coating (e.g., spin-on coating), soft baking, mask aligning, exposure, post-exposure baking, developing the photoresist, rinsing and drying (e.g., hard baking). The etching process may include a dry etching process or a wet etching process.

(22) After the fin structures **104** are formed, a liner layer may be formed over the fin structures **104** and in the trenches between the fin structures **104**. The liner layer may be conformally formed over the substrate **102**, the fin structure **110**, and the mask structure covering the fin structure **110**. The liner layer may be used to protect the fin structure **110** from being damaged in the following processes (such as an anneal process or an etching process). The liner layer may be made of silicon

nitride. The liner layer may be formed by using a thermal oxidation, a CVD process, an atomic layer deposition (ALD) process, a LPCVD process, a plasma enhanced CVD (PECVD) process, a HDPCVD process, a flowable CVD (FCVD) process, another applicable process, or a combination thereof.

(23) Next, an isolation structure material **112** is then filled into the trenches between the fin structures **104** and over the liner layer, as shown in FIG. **1**C in accordance with some embodiments. The isolation structure material **112** may be made of silicon oxide, silicon nitride, silicon oxynitride (SiON), fluoride-doped silicate glass (FSG), other low-k dielectric materials, or a combination thereof. The isolation structure material **112** may be deposited using a deposition process, such as a chemical vapor deposition (CVD) process (e.g. a flowable CVD (FCVD) process), a spin-on-glass process, or another applicable process.

(24) Next, the isolation structure material **112** and the liner layer are etched back using an etching process, and an isolation structure **112** is formed surrounding the base fin structure, as shown in FIG. **1**C in accordance with some embodiments. The etching process may be used to remove the top portion of the liner layer and the top portion of the isolation structure material **112**. As a result, the first semiconductor material layers **106** and the second semiconductor material layers **108** may be exposed. The isolation structure **112** may be a shallow trench isolation (STI) structure. The isolation structure **112** may be configured to electrically isolate active regions such as fin structures **104** of the semiconductor structure **10***a* and prevent electrical interference and crosstalk.

(25) Next, a dummy gate structure **116** is formed over and across the fin structures **104**, as shown in FIG. **1**D in accordance with some embodiments. The dummy gate structure **116** may be used to define the source/drain regions and the channel regions of the resulting semiconductor structure **10***a*. The dummy gate structure **116** may include a dummy gate dielectric layer **118** and a dummy gate electrode layer **120**. The dummy gate dielectric layer **118** and the dummy gate electrode layer **120** may be replaced by the following steps to form a real gate structure with a high-k dielectric layer and a metal gate electrode layer.

(26) The dummy gate dielectric layer **118** may include one or more dielectric materials, such as silicon oxide, silicon nitride, silicon oxynitride (SiON), HfO_2, HfZrO, HfSiO, HfTiO, HfAlO, or a combination thereof. The dummy gate dielectric layer **118** may be formed by an oxidation process (e.g., a dry oxidation process, or a wet oxidation process), a chemical vapor deposition process, other applicable processes, or a combination thereof. Alternatively, the dummy gate dielectric layer **118** may include a high-k dielectric layer (e.g., the dielectric constant is greater than 3.9) such as hafnium oxide (HfO_2). Alternatively, the high-k dielectric layer may include other high-k dielectrics, such as LaO, AlO, ZrO, TiO, Ta_2O_5, Y_2O_3, $SrTiO_3$, $BaTiO_3$, BaZrO, HfZrO, HfLaO, HfTaO, HfSiO, HfSiON, HfTiO, LaSiO, AlSiO, $(Ba, Sr)TiO_3$, Al_2O_3, other applicable high-k dielectric materials, or a combination thereof. The high-k dielectric layer may be formed by a chemical vapor deposition process (e.g., a plasma enhanced chemical vapor deposition (PECVD) process, or a metalorganic chemical vapor deposition (MOCVD) process), an atomic layer deposition (ALD) process (e.g., a plasma enhanced atomic layer deposition (PEALD) process), a physical vapor deposition (PVD) process (e.g., a vacuum evaporation process, or a sputtering process), other applicable processes, or a combination thereof.

(27) The dummy gate electrode layer **120** may include polycrystalline-silicon (poly-Si), poly-crystalline silicon-germanium (poly-SiGe), other applicable materials, or a combination thereof. The dummy gate electrode layer **120** may be formed by a chemical vapor deposition process (e.g., a low pressure chemical vapor deposition process, or a plasma enhanced chemical vapor deposition process), a physical vapor deposition process (e.g., a vacuum evaporation process, or a sputtering process), other applicable processes, or a combination thereof.

(28) Hard mask layers **122** are formed over the dummy gate structure **116**, as shown in FIG. **1**D in accordance with some embodiments. The hard mask layers **122** may include multiple layers, such

as an oxide layer **124** and a nitride layer **126**. The oxide layer **124** may include silicon oxide, and the nitride layer **126** may include silicon nitride.
(29) The formation of the dummy gate structure **116** may include conformally forming a dielectric material as the dummy gate dielectric layer **118**. Afterwards, a conductive material may be formed over the dielectric material as the dummy gate electrode layers **120**, and the bi-layered hard mask layers **122**, including the oxide layer **124** and the nitride layer **126**, may be formed over the conductive material. Next, the dielectric material and the conductive material may be patterned and etched through the bi-layered hard mask layers **122** to form the dummy gate structure **116**, as shown in FIG. **1**D in accordance with some embodiments. The dummy gate dielectric layer **118** and the dummy gate electrode layer **120** may be etched by a dry etching process. After the etching process, the first semiconductor material layers **106** and the second semiconductor material layers **108** may be exposed on opposite sides of the dummy gate structure **116**.
(30) Next, a conformal dielectric layer is formed over the substrate **102** and the dummy gate structure **116**, and then an etching process is performed. A pair of spacer layers **128** is formed over opposite sidewalls of the dummy gate structure **116**, and a source/drain opening is formed beside the dummy gate structure **116**, as shown in FIG. **1**E in accordance with some embodiments.
(31) The spacer layers **128** may be multi-layer structures formed by different materials with different etching selectivity. The spacer layers **128** may be made of silicon oxide, silicon nitride, silicon oxynitride, and/or dielectric materials. The spacer layers **128** may be formed by a chemical vapor deposition (CVD) process, a spin-on-glass process, or another applicable process.
(32) After the spacer layers **128** are formed, the first semiconductor material layers **106** and the second semiconductor material layers **108** of the fin structure **110** not covered by the dummy gate structure **116** and the spacer layers **128** are etched to form the trenches beside the dummy gate structure **116**, as shown in FIG. **1**F in accordance with some embodiments.
(33) The fin structures **104** may be recessed by performing a number of etching processes. That is, the first semiconductor material layers **106** and the second semiconductor material layers **108** of the fin structures **104** may be etched in different etching processes. The etching process may be a dry etching process or a wet etching process. The fin structures **104** may be etched by a dry etching process.
(34) Next, the first semiconductor material layers **106** are laterally etched from the source/drain opening to form recesses, as shown in FIG. **1**G in accordance with some embodiments. The outer portions of the first semiconductor material layers **106** may be removed, and the inner portions of the first semiconductor material layers **106** under the dummy gate structure **116** and the spacer layers **128** may remain. After the lateral etching process, the sidewalls of the etched first semiconductor material layers **106** may be not aligned with the sidewalls of the second semiconductor material layers **108**.
(35) The lateral etching of the first semiconductor material layers **106** may be a dry etching process, a wet etching process, or a combination thereof. In some embodiments, the first semiconductor material layers **106** are Ge or SiGe and the second semiconductor material layers **108** are Si, and the first semiconductor material layers **106** are selectively etched to form the recesses by using a wet etchant such as, but not limited to, ammonium hydroxide (NH_4OH), tetramethylammonium hydroxide (TMAH), ethylenediamine pyrocatechol (EDP), or potassium hydroxide (KOH) solutions, or the like.
(36) Next, an inner spacer **134** is formed in the recess, as shown in FIG. **1**H in accordance with some embodiments. The inner spacer **134** may provide a barrier between subsequently formed source/drain epitaxial structures and gate structure. The inner spacer **134** may be made of dielectric material such as silicon oxide (SiO_2), silicon nitride (SiN), silicon carbide (SiC), silicon oxynitride (SiON), silicon carbon nitride (SiCN), silicon oxide carbonitride (SiOCN), or a combination thereof. The inner spacer **134** may be formed using a deposition process. The deposition process may include a CVD process (such as LPCVD, PECVD, SACVD, or FCVD), an

ALD process, another applicable method, or a combination thereof.
(37) Next, a source/drain epitaxial structure **140** is formed in the source/drain opening, as shown in FIG. **1**I in accordance with some embodiments. The source/drain epitaxial structure **140** may be formed over opposite sides of the dummy gate structure **116**. Source/drain epitaxial structure **140** may refer to a source or a drain, individually or collectively dependent upon the context.
(38) A strained material may be grown in the source/drain opening using an epitaxial (epi) process to form the source/drain epitaxial structure **140**. The lattice constant of the strained material may be different from the lattice constant of the substrate **102**. The source/drain epitaxial structure **140** may include SiGeB, SiP, SiAs, SiGe, other applicable materials, or a combination thereof. The source/drain epitaxial structure **140** may be formed by an epitaxial growth step, such as metalorganic chemical vapor deposition (MOCVD), metalorganic vapor phase epitaxy (MOVPE), plasma-enhanced chemical vapor deposition (PECVD), remote plasma-enhanced chemical vapor deposition (RP-CVD), molecular beam epitaxy (MBE), hydride vapor phase epitaxy (HYPE), liquid phase epitaxy (LPE), chloride vapor phase epitaxy (Cl-VPE), or any other suitable method.
(39) The source/drain epitaxial structure **140** may be in-situ doped during the epitaxial growth process. For example, the source/drain epitaxial structures **140** may be the epitaxially grown SiGe doped with boron (B). For example, the source/drain epitaxial structure **140** may be the epitaxially grown Si doped with carbon to form silicon:carbon (Si:C) source/drain features, phosphorous to form silicon:phosphor (Si:P) source/drain features, or both carbon and phosphorous to form silicon carbon phosphor (SiCP) source/drain features. The source/drain epitaxial structure **140** may be doped in one or more implantation processes after the epitaxial growth process.
(40) Next, a contact etch stop layer **142** is formed over the source/drain epitaxial structure **140**, as shown in FIG. **1**J in accordance with some embodiments. More specifically, the contact etch stop layer **142** covers the sidewalls of the spacer layers **128** and the source/drain epitaxial structures **140** in accordance with some embodiments.
(41) The contact etch stop layer **142** may be made of a dielectric material such as silicon nitride, silicon oxide, silicon oxynitride (SiON), other applicable materials, or a combination thereof. The contact etch stop layer **142** may be formed by a chemical vapor deposition process (e.g., a plasma enhanced chemical vapor deposition (PECVD) process, or a metalorganic chemical vapor deposition (MOCVD) process), an atomic layer deposition (ALD) process (e.g., a plasma enhanced atomic layer deposition (PEALD) process), a physical vapor deposition (PVD) process (e.g., a vacuum evaporation process, or a sputtering process), other applicable processes, or a combination thereof.
(42) After the contact etch stop layer **142** is formed, an inter-layer dielectric (ILD) structure **144** is formed over the contact etch stop layer **142**, as shown in FIG. **1**J in accordance with some embodiments. The ILD structure **144** may include multilayers made of multiple dielectric materials, such as silicon oxide (SiO_x, where x may be a positive integer), silicon oxycarbide ($SiCO_y$, where y may be a positive integer), silicon oxycarbonitride ($SiNCO_z$, where z may be a positive integer), silicon nitride, silicon oxynitride, phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), low-k dielectric material, or another applicable dielectric material. Examples of low-k dielectric materials include, but are not limited to, fluorinated silica glass (FSG), carbon doped silicon oxide, amorphous fluorinated carbon, parylene, bis-benzocyclobutenes (BCB), or polyimide. The ILD structure **144** may be formed by chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), spin-on coating, or another applicable process.
(43) Afterwards, a planarizing process or an etch-back process is performed on the ILD structure **144** until the top surface of the dummy gate structure **116** is exposed, as shown in FIG. **1**J in accordance with some embodiments. After the planarizing process, the top surface of the dummy gate structure **116** may be substantially level with the top surfaces of the spacer layers **128** and the ILD structure **144**. The hard mask layers **122** including the oxide layer **124** and the nitride layer

126 may be removed during the planarizing process. The planarizing process may include a grinding process, a chemical mechanical polishing (CMP) process, an etching process, other applicable processes, or a combination thereof.
(44) Next, the dummy gate structure **116** is removed, as shown in FIG. **1**K in accordance with some embodiments. Therefore, a trench **146** is formed between the spacer layers **128** over the fin structure **110** and the first semiconductor material layers **106** may be exposed from the trench **146**.
(45) The dummy gate structure **116** may be removed by a dry etching process or a wet etching process. The removal process may include one or more etching processes. For example, when the dummy gate electrode layers **120** are polysilicon, a wet etchant such as a tetramethylammonium hydroxide (TMAH) solution may be used to selectively remove the dummy gate electrode layers **120**. Afterwards, the dummy gate dielectric layer **118** may be removed using a plasma dry etching, a dry chemical etching, and/or a wet etching.
(46) Next, the first semiconductor material layers **106** are removed and gaps are formed between the first semiconductor material layers **106**, as shown in FIG. **1**K in accordance with some embodiments. More specifically, the second semiconductor material layers **108** exposed by the gaps form nanostructures **108**, and the nanostructures **108** are configured to function as channel regions in the resulting semiconductor devices in accordance with some embodiments.
(47) The first semiconductor material layers **106** may be removed by performing one or more etching processes. The etching process may include a selective wet etching process, such as APM (e.g., ammonia hydroxide-hydrogen peroxide-water mixture) etching process. The wet etching process uses etchants such as ammonium hydroxide (NH_4OH), TMAH, ethylenediamine pyrocatechol (EDP), and/or potassium hydroxide (KOH) solutions.
(48) Next, a gate structures **148** are formed surrounding the nanostructures **108** and over the nanostructures **108**. Gate structures **148** are formed surrounding the nanostructures **108** to form gate-all-around (GAA) transistor structures, as shown in FIG. **1**L in accordance with some embodiments. Therefore, the gate control ability may be enhanced.
(49) In some embodiments as shown in FIG. **1**L, the gate structures **148** are multi-layered structures. Each of the gate structures **148** may include an interfacial layer, a gate dielectric layer **150**, a work function layer **152**, and a gate electrode layer, as shown in FIG. **1**L in accordance with some embodiments. For the purpose of brevity, only the gate dielectric layer **150** and the work function layer **152** are shown in FIG. **1**L.
(50) The interfacial layer may be formed around the nanostructures **108**. The interfacial layer may be made of silicon oxide, and the interfacial layer may be formed by thermal oxidation.
(51) The gate dielectric layer **150** is formed over the interfacial layer, so that the nanostructures **108** are surrounded (e.g. wrapped) by the gate dielectric layer **150**. In addition, the gate dielectric layer **150** also covers the sidewalls of the spacer layers **128** and the inner spacers **134** in accordance with some embodiments. The gate dielectric layer **150** may be made of one or more layers of dielectric materials, such as HfO_2, HfSiO, HfSiON, HfTaO, HfTiO, HfZrO, zirconium oxide, aluminum oxide, titanium oxide, hafnium dioxide-alumina (HfO_2—Al_2O_3) alloy, other applicable high-k dielectric materials, or a combination thereof. The gate dielectric layer **150** may be formed using CVD, ALD, other applicable methods, or a combination thereof.
(52) The work function layer **152** may be conformally formed surrounding the nanostructures **108**. The work function layer **152** may be also formed over the nanostructures **108**. The work function layer **152** may be multi-layer structures.
(53) The work function layer **152** may be made of metal materials. The metal materials of the work function layer **152** may include N-work-function metal. The N-work-function metal may include tungsten (W), copper (Cu), titanium (Ti), silver (Ag), aluminum (Al), titanium aluminum alloy (TiAl), titanium aluminum nitride (TiAlN), tantalum carbide (TaC), tantalum carbon nitride (TaCN), tantalum silicon nitride (TaSiN), manganese (Mn), zirconium (Zr), or a combination thereof. The work function layer **152** may include P-work-function metal. The P-work-function

metal may include titanium nitride (TiN), tungsten nitride (WN), tantalum nitride (TaN), ruthenium (Ru) or a combination thereof. The work function layer **152** may be formed by using CVD, ALD, another applicable method, or a combination thereof.

(54) The gate electrode layer may be formed over the work function layer **152**. The gate electrode layer may be made of one or more layers of conductive material, such as aluminum, copper, titanium, tantalum, tungsten, cobalt, molybdenum, tantalum nitride, nickel silicide, cobalt silicide, TiN, WN, TiAl, TiAlN, TaCN, TaC, TaSiN, metal alloys, another suitable material, or a combination thereof. The gate electrode layer may be formed using CVD, ALD, electroplating, another applicable method, or a combination thereof. After the gate electrode layer is formed, a planarization process such as CMP or an etch-back process may be performed.

(55) Next, a source/drain opening is formed in the ILD structure **144** to expose the source/drain epitaxial structure **140**, and a silicide layer **161** may be formed over the source/drain epitaxial structure **140**. The silicide layer **161** may reduce the contact resistance between the source/drain epitaxial structure **140** and the subsequently formed source/drain contact structure over the source/drain epitaxial structure **140**. The silicide layer **161** may be made of titanium silicide (TiSi.sub.2), nickel silicide (NiSi), cobalt silicide (CoSi), or other suitable low-resistance materials.

(56) The silicide layer **161** may be formed over the source/drain epitaxial structure **140** by forming a metal layer over the source/drain epitaxial structure **140** first. The metal layer may react with the source/drain epitaxial structure **140** in an annealing process and a silicide layer **161** may be produced. Afterwards, the unreacted metal layer may be removed in an etching process and the silicide layer **161** may be left.

(57) Next, a barrier layer (not shown) may be conformally formed over the bottom surface and the sidewalls of the source/drain opening. Afterwards, the barrier layer may be etched back. The barrier layer remains over the bottom surface of the source/drain opening. The barrier layer may be formed before filling the conductive material in the source/drain opening to prevent the conductive material from diffusing out. The barrier layer may also serve as an adhesive or glue layer. The material of the barrier layer may be tantalum, titanium, titanium nitride, other applicable materials, or a combination thereof. The barrier layer may be formed by depositing the barrier layer materials by a physical vapor deposition process (PVD) (e.g., evaporation or sputtering), an atomic layer deposition process (ALD), an electroplating process, other applicable processes, or a combination thereof.

(58) Afterwards, a source/drain contact structure **162** is formed into the source/drain opening over the source/drain epitaxial structure **140**, as shown in FIG. **1**M in accordance with some embodiments. The source/drain contact structure **162** may be partially land or fully land over the source/drain epitaxial structure **140**.

(59) The source/drain contact structure **162** may be made of aluminum (Al), copper (Cu), tungsten (W), titanium (Ti), tantalum (Ta), titanium nitride (TiN), cobalt, tantalum nitride (TaN), nickel silicide (NiS), cobalt silicide (CoSi), copper silicide, tantalum carbide (TaC), tantalum silicide nitride (TaSiN), tantalum carbide nitride (TaCN), titanium aluminide (TiAl), titanium aluminum nitride (TiAlN), other applicable conductive materials, or a combination thereof.

(60) The source/drain contact structure **162** may be formed by a chemical vapor deposition process (CVD), a physical vapor deposition process (PVD), (e.g., evaporation or sputter), an atomic layer deposition process (ALD), a plasma enhanced CVD (PECVD), a plasma enhanced physical vapor deposition (PEPVD), an electroplating process, another suitable process, or a combination thereof to deposit the conductive materials of the source/drain contact structure **162**, and then a planarization process such as a chemical mechanical polishing (CMP) process or an etch back process is optionally performed to remove excess conductive materials. After the planarization process, the top surface of the source/drain contact structure **162** may be level with the top surfaces of the ILD structure **144** and the spacer layers **128**.

(61) Next, after the front side S/D contact structure **162** are formed, a front end structure **164** is

formed over the gate structures **148**, the ILD layer **144**, and the front side S/D contact structure **162**, as shown in FIG. **1**M in accordance with some embodiments. The front end structure **164** may include an etch stop layer and various features (not shown), such as a multilayer interconnect structure (such as contacts to gate, vias, lines, inter metal dielectric layers, passivation layers, etc.) formed thereon.

(62) Next, as shown in FIG. **1**N in accordance with some embodiments, after the front end structure **164** is formed, a carrier substrate (not shown) is attached to the front end structure **164**, and then the substrate **102** is flipped. Afterwards, a planarization is performed on the back side of the substrate **102**. More specifically, a planarization is performed on the substrate **102** until the epitaxial sacrificial structures **136** and the CESL **142** are exposed.

(63) The planarization process may be an etching process, a CMP process, a mechanical grinding process, a dry polishing process, or a combination thereof. The front end structure **164** is configured to support the semiconductor structure in subsequent manufacturing process.

(64) It is appreciated that although the structures in FIG. **1**N is shown in upside down for better understanding the manufacturing processes, the spatial positions of the elements (e.g. top portions, bottom portions, topmost, bottommost, or the like) are described according to the original positions shown in FIGS. **1**A to **1**M so they can be in consistence with those described previously for clarity. For example, the front side surface of the S/D epitaxial structure **140** refers to the surface that is in contact with the S/D contact structure **162**, and the back side surface of the S/D structures **140** refers to the surface that is in contact with the substrate **102**, since the structure shown in FIG. **1**N is upside down.

(65) FIGS. **1**O-**1**W show cross-sectional representations of various stages of manufacturing the semiconductor structure **10*a*** after FIG. **1**N, in accordance with some embodiments.

(66) After the substrate **102** is flipped, a hard mask layer **172** may be formed over the back side of the substrate **102**. The hard mask layer **172** may be made of silicon nitride, silicon carbon nitride (SiCN), or applicable material. The hard mask layer **172** may be formed by a deposition process, such as low-pressure CVD (LPCVD) process, plasma enhanced CVD (PECVD) process, or another deposition process.

(67) Afterwards, a portion of the hard mask layer **172**, the substrate **102**, and the source/drain epitaxial structure **140** are removed, and a trench **175** is formed, as shown in FIG. **1**O, in accordance with some embodiments. The trench **175** may be formed by a dry etching process or a wet etching process. The source/drain epitaxial structure **140** may have a concave bottom surface after the trench **175** is formed.

(68) Afterwards, after the trench **175** is formed, a liner layer **168** is formed over the sidewalls of the trench **175**, as shown in FIG. **1**P in accordance with some embodiments. The liner layer **168** may be made of SiN, SiCN, SiOCN, SiON, other applicable materials, or a combination thereof. In some embodiments, the liner layer **168** is made of SiN.

(69) In some embodiments, the liner layer **168** has a thickness in a range of about 2 nm to about 5 nm. If the liner layer **168** is too thick, the area of subsequently formed conductive material may be too small, and it may be difficult to fill the conductive material in the trench **175**. If the liner layer **168** is too thin, the liner layer **168** may not be uniformly formed over the sidewalls of the trench **175**.

(70) The liner layer **168** may be conformally formed over sidewalls and bottom surface of the trench **175** and over the top surface of the hard mask layer **172** first, and then the liner layer **168** over the bottom surface of the trench **175** and over the top surface of the hard mask layer **172** may be removed. The liner layer **168** may be formed by a deposition process, such as chemical vapor deposition (CVD), physical vapor deposition, (PVD), atomic layer deposition (ALD), or another applicable process. The liner layer **168** may be removed by a dry etching process. The liner layer **168** may increase the isolation between the subsequently formed conductive material and the gate structure **148**.

(71) Next, a conductive material layer is formed in the trench **175** and over the liner layer **168** and the hard mask layer **172** (not shown). The conductive material layer may be formed by a deposition process, such as chemical vapor deposition (CVD), physical vapor deposition, (PVD), atomic layer deposition (ALD), or another applicable process.
(72) Afterwards, a portion of the conductive material layer is annealed to form a first silicide layer **180** on the exposed S/D structure **140** by an annealing process, as shown in FIG. **1**Q in accordance with some embodiments. The first silicide layer **180** is in direct contact with the S/D structure **140** and the liner layer **168**. The first silicide layer **180** is formed by annealing the conductive material layer so the conductive material layer reacts with the S/D structures **140** to form the first silicide layer **180**. The first silicide layer **180** may be made of NiSi, TiNiSi, CoSi, MoSi, RuSi, TiSi, or the like. In some embodiments, the first silicide layer **180** is made of TiSi.
(73) In some embodiments, the first silicide layer **180** has a thickness in a range of about 0.5 nm to about 10 nm. If the first silicide layer **180** is too thin, the resistance may be worse.
(74) Next, a nitrogen treatment process may be performed on the conductive material layer after the annealing process to form a second silicide layer **182**, as shown in FIG. **1**Q in accordance with some embodiments. The conductive material layer **180** is nitridized by the nitrogen treatment process.
(75) After the nitrogen treatment process, the conductive material layer becomes the second silicide layer **182**. The second silicide layer **182** is made of a nitrogen-containing compound or is made of nitrogen. The second silicide layer **182** may be made of NiSiN, TiNiSiN, CoSiN, MoSiN, RuSiN, TiSiN. In some embodiments, the second silicide layer **182** is made of TiSiN. In some embodiments, the first silicide layer **180** and the second silicide layer **182** include different materials.
(76) Next, a first conductive layer **186** is formed in the trench **175** over the second silicide layer **182**, as shown in FIG. **1**R, in accordance with some embodiments. The first conductive layer **186** may be made of Co, Moly, Cu, Ru, W, other applicable materials, or a combination thereof. In some embodiments, the first conductive layer **186** is made of Ru.
(77) The first conductive layer **186** may be formed by a bottom-up deposition process, which is formed form bottom to top. The first silicide layer **180** is located at bottom to help the formation of the first conductive layer **186**. By using the bottom-up deposition process, there may be no glue layer formed between the second silicide layer **182** and the first conductive layer **186**. The resistance of a glue layer is higher than that of the first conductive layer **186**. Therefore, the resistance may be decreased without the glue layer. Therefore, the performance of the semiconductor structure **10***a* may be improved.
(78) In some embodiments, the first conductive layer **186** is thicker at the bottom of the trench **175** than over the hard mask layer **172**.
(79) Next, the first conductive layer **186** is etched back, and the first conductive layer **186** over the upper sidewalls of the trench **175** and over the hard mask layer **172** is removed, as shown in FIG. **1***s*, in accordance with some embodiments. After etching back the first conductive layer **186**, the first conductive layer **186** has a dishing shape top surface. The first conductive layer **186** may have a concave top surface.
(80) In some embodiments, the first conductive layer **186** has a height at the bottom of the trench **175** in a range of about 3 nm to about 30 nm. If the first conductive layer **186** is too short, the first silicide layer **180** and the second silicide layer **182** may be damaged in the following etching process. If the first conductive layer **186** is too high, the resistance may be increased.
(81) The first conductive layer **186** may be etched back by chemicals including Cl.sub.2 and O.sub.2. The ratio of Cl.sub.2 and O.sub.2 is in a range of about 1:10 to about 10:1. The first conductive layer **186** may be etched back by chemicals including F.sub.2. Afterwards, a H.sub.2 treatment may be optionally performed.
(82) Afterwards, the second silicide layer **182** is etched back, and the upper sidewalls of the liner

layer **168** and the top surface of the hard mask layer **172** are exposed, as shown in FIG. **1**T in accordance with some embodiments. After etching back the second silicide layer **182**, the second silicide layer **182** has a dishing shape top surface. The second silicide layer **182** may have a concave top surface. In some embodiments, the top surface of the first conductive layer **186** is more concave than the top surface of the second silicide layer **182**.
(83) In some embodiments, the dishing amount of the second silicide layer **182** is in a range of about 0.5 nm to about 3 nm.
(84) In some embodiments, the second silicide layer **182** surrounds the first conductive layer **186**. In some embodiments, second silicide layer **182** is formed at the bottom surface and the sidewalls of the first conductive layer **186**.
(85) In some embodiments, the second silicide layer **182** has a thickness in a range of about 0.5 nm to about 10 nm. If the second silicide layer **182** is too thick, the area of the first conductive layer **186** may be too small, and the resistance may be increased. If the second silicide layer **182** is too thin, the resistance may be increased.
(86) The second silicide layer **182** may be etched back by chemicals including DI—O.sub.3 water. The second silicide layer **182** may be etched back under a temperature in a range of about 20° C. to about 100° C. The second silicide layer **182** may be etched back for about seconds to about 100 seconds.
(87) The second silicide layer **182** may be also etched back by an O.sub.2 treatment and followed by an HCl wet etching process. The **02** treatment may be performed under a temperature in a range of about 100° C. to about 250° C. with an O.sub.2 flow of about 1000 sccm to about 5000 sccm. The HCl wet etching process may be performed under a temperature in a range of about 20° C. to about 100° C. The HCl wet etching process may be performed for about 30 seconds to about 100 seconds.
(88) In some embodiments, the second silicide layer **182** and the first conductive layer **186** may be removed by different chemicals.
(89) Next, a second conductive layer **190** is formed in the trench **175** over the first conductive layer **186**, as shown in FIG. **1**U in accordance with some embodiments. The second conductive layer **190** protrudes over the hard mask layer **172**. In some embodiments, the second conductive layer **190** has a curved convex upper surface. In some embodiments, the second conductive layer **190** is in direct contact with the liner layer **168**.
(90) The second conductive layer **190** may be made of Co, Moly, Cu, Ru, W, other applicable materials, or a combination thereof. In some embodiments, the second conductive layer **190** is made of W. In some embodiments, the first conductive layer **186** and the second conductive layer **190** are made of different materials. In some embodiments, the first conductive layer **186** and the second conductive layer **190** are made of the same material.
(91) In some embodiments, the second conductive layer **190** has a thickness in a range of about 3 nm to about 50 nm. If the second conductive layer **190** is too short, the first silicide layer **180** and the second silicide layer **182** may be damaged in the following etching process. If the second conductive layer **190** is too high, the resistance may be increased.
(92) The second conductive layer **190** may be formed by a bottom-up deposition process, which is formed form bottom to top. By using the bottom-up deposition process, there may be no glue layer formed between the liner layer **168** and the second conductive layer **190**. The resistance of a glue layer is higher than that of the second conductive layer **190**. Therefore, the resistance may be decreased without the glue layer. Therefore, the performance of the semiconductor structure **10***a* may be improved by using the bottom-up deposition process.
(93) Next, a overburden layer **192** is deposited over the hard mask layer **172** and the second conductive layer **190**, as shown in FIG. **1**V in accordance with some embodiments. The overburden layer **192** may provide a uniform level for subsequently planarization process. The overburden layer **192** may include SiN, SiO, SiON, SiCN, other applicable materials, or a combination thereof.

The overburden layer **192** may be formed by chemical vapor deposition (CVD), physical vapor deposition, (PVD), atomic layer deposition (ALD), or another applicable process.
(94) Afterwards, a planarization process may be performed, and the overburden layer **192** and a portion of the hard mask layer **172** may be removed, and a back-side source/drain contact structure **194** is formed, as shown in FIG. **1**W in accordance with some embodiments. The planarization process may be an etching process, a CMP process, a mechanical grinding process, a dry polishing process, or a combination thereof. In some embodiments, the liner layer **168** is formed over sidewalls of the back-side source/drain contact structure **194**.
(95) FIG. **2** is an enlarged cross-sectional representation of a semiconductor device structure **10***a*, in accordance with some embodiments of the disclosure. In some embodiments, both the first conductive material **186** and the second silicide layer **182** have concave top surfaces.
(96) By forming the back-side source/drain contact structure **194** by two-step etching back process, the back-side source/drain contact structure **194** may be formed by bottom-up deposition process without void or selective loss. In addition, the first silicide layer **180** may not be damaged, and the contact resistance may be decreased.
(97) Many variations and/or modifications may be made to the embodiments of the disclosure. FIGS. **3**A-**3**B are perspective representations of various stages of forming a semiconductor device structure **10***b*, in accordance with some embodiments of the disclosure. Some processes or devices are the same as, or similar to, those described in the embodiments above, and therefore the descriptions of these processes and devices are not repeated herein. The difference from the embodiments described above is that, as shown in FIG. **3**A in accordance with some embodiments, an implantation process **300** is performed after forming the second conductive layer **190**.
(98) In some embodiments, the implantation process **300** includes using a germanium (Ge)-containing compound. The adhesion between the liner layer **168** and the second conductive layer **190** may be improved by the Ge-containing compound.
(99) In some embodiments, the implantation process **300** is performed by using a fluorine (F)-containing compound or a carbon (C)-containing compound. The dielectric constant may be reduced by the F-containing compound or the C-containing compound. The capacitance and the performance may be further improved.
(100) In some embodiments, the Ge-containing compound is implanted at an energy in a range from about 30 keV to about 50 keV. The Ge-containing compound may be distributed in the substrate **102** and the hard mask layer **172**. In some embodiments, the dosage of the Ge-containing compound is from about 1E13(cm.sup.−3) to about 1E16(cm.sup.−3). In some embodiments, the tilt angle of the implantation process **300** is in a range from about 20 degree to about 50 degree.
(101) In some embodiments, the F-containing compound or the C-containing compound are implanted at an energy in a range from about 20 keV to about 40 keV. The F-containing compound or the C-containing compound may be distributed in the hard mask layer **172**. In some embodiments, the dosage of the F-containing compound or the C-containing compound is from about 1E13(cm.sup.−3) to about 1E16(cm.sup.−3). In some embodiments, the tilt angle of the implantation process **300** is in a range from about 20 degree to about 50 degree.
(102) Afterwards, an overburden layer **192** is formed and a planarization process is performed to form the back-side source/drain contact structure **194**, as shown in FIG. **3**B in accordance with some embodiments. After the implantation process **300**, the Ge-containing compound may be distributed in the hard mask layer **172**, the substrate **102**, and the back-side source/drain contact structure **194**.
(103) By forming the back-side source/drain contact structure **194** by two-step etching back process, the back-side source/drain contact structure **194** may be formed by bottom-up deposition process without void or selective loss. In addition, the first silicide layer **180** may not be damaged, and the contact resistance may be decreased. An implantation process performed with the Ge-containing compound after depositing the second conductive layer **190** may improve the adhesion

between the liner layer **168** and the second conductive layer **190**. An implantation process performed with the F-containing compound or C-containing compound after depositing the second conductive layer **190** may lower the capacitance.
(104) Many variations and/or modifications may be made to the embodiments of the disclosure. FIGS. **4**A-**4**B are perspective representations of various stages of forming a semiconductor device structure **10***c*, in accordance with some embodiments of the disclosure. Some processes or devices are the same as, or similar to, those described in the embodiments above, and therefore the descriptions of these processes and devices are not repeated herein. The difference from the embodiments described above is that, as shown in FIG. **4**A in accordance with some embodiments, a barrier layer **196** is formed before depositing the first conductive layer **186**.
(105) In some embodiments, the first conductive layer **186** is formed by CVD such as low-pressure CVD (LPCVD) process, plasma enhanced CVD (PECVD) process, or another deposition process. Therefore, the barrier layer **196** needs to be formed to enhance adhesion between the second silicide layer **182** and the first conductive layer **186**. The barrier layer **196** may also prevent the first conductive layer **186** from diffusing out.
(106) Afterwards, the barrier layer **196** is etched back while etching back the first conductive layer **186**, as shown in FIG. **4**B in accordance with some embodiments. In some embodiments, after the etching back process, the top surface of the barrier layer **196** is higher than the top surface of the first conductive layer **186**.
(107) Next, the second silicide layer **182** is etched back, and the top surface of the second silicide layer **182** has a dishing shape, as shown in FIG. **4**C in accordance with some embodiments. The processes for etching back the second silicide layer **182** may be the same as, or similar to, those used to etch back the second silicide layer **182** in the previous embodiments. For the purpose of brevity, the descriptions of these processes are not repeated herein.
(108) Afterwards, a second conductive layer **190** is formed over the first conductive layer **186**, the barrier layer **196**, and the second silicide layer **182**, and a planarization process is performed to form the back-side source/drain contact structure **194**, as shown in FIG. **4**C in accordance with some embodiments.
(109) By forming the back-side source/drain contact structure **194** by two-step etching back process, the back-side source/drain contact structure **194** may be formed by bottom-up deposition process without void or selective loss. In addition, the first silicide layer **180** may not be damaged, and the contact resistance may be decreased. A barrier layer **196** may be formed between the second silicide layer **182** and the first conductive layer **186** while the first conductive layer **186** is formed by a CVD deposition process.
(110) Many variations and/or modifications may be made to the embodiments of the disclosure. FIG. **5** is an enlarged cross-sectional representation of a semiconductor device structure **10***d*, in accordance with some embodiments of the disclosure. Some processes or devices are the same as, or similar to, those described in the embodiments above, and therefore the descriptions of these processes and devices are not repeated herein. The difference from the embodiments described above is that, as shown in FIG. **5** in accordance with some embodiments, the second silicide layer **182** has a flat top surface.
(111) When etching back the second silicide layer **182**, if the etching time is shorter, the top surface of the second silicide layer **182** may be flat.
(112) By forming the back-side source/drain contact structure **194** by two-step etching back process, the back-side source/drain contact structure **194** may be formed by bottom-up deposition process without void or selective loss. In addition, the first silicide layer **180** may not be damaged, and the contact resistance may be decreased. The second silicide layer **182** may have a flat top surface by modifying the etching time.
(113) Many variations and/or modifications may be made to the embodiments of the disclosure. FIG. **6** is an enlarged cross-sectional representation of a semiconductor device structure **10***e*, in

accordance with some embodiments of the disclosure. Some processes or devices are the same as, or similar to, those described in the embodiments above, and therefore the descriptions of these processes and devices are not repeated herein. The difference from the embodiments described above is that, as shown in FIG. **6** in accordance with some embodiments, the top surface of the first conductive layer **186** is higher than the top surface of the second silicide layer **182**.
(114) When etching back the first conductive layer **186**, if the etching time is shorter, the top surface of the first conductive layer **186** may be flat. The shape of the top surfaces of the first conductive layer **186** and the second silicide layer **182** may be modified independently by the etching back processes of the first conductive layer **186** and the second silicide layer **182**.
(115) By forming the back-side source/drain contact structure **194** by two-step etching back process, the back-side source/drain contact structure **194** may be formed by bottom-up deposition process without void or selective loss. In addition, the first silicide layer **180** may not be damaged, and the contact resistance may be decreased. The first conductive layer **186** may have a flat top surface by modifying the etching time. The top surfaces of each of the first conductive layer **186** and the second silicide layer **182** may be independently modified by the etching time of different etching processes.
(116) As described previously, the back-side source/drain contact structure **194** by two-step etching back process. The resistance may be reduced by using bottom-up deposition process. The two-step etching back process may help to reduce the resistance and defects when forming the bottom-up deposited conductive layer. The shape of the back-side source/drain contact structure **194** may be modified by the process parameter of the two-step etching back process. In some embodiments as shown in FIG. **3**A, and implantation process **300** is performed, and the adhesion between the liner layer **168** and the second conductive layer **190** is improved. In some embodiments as shown in FIG. **4**A, a barrier layer **196** is formed under the first conductive layer **186** when the first conductive layer is formed by a CVD deposition process. In some embodiments as shown in FIG. **5**, the second silicide layer **182** has a flat top surface. In some embodiments as shown in FIG. **6**, the top surface of the first conductive layer **186** is higher than the top surface of the second silicide layer **182**, and the top surfaces of the first conductive layer **186** and the second silicide layer **182** can be modified independently.
(117) Embodiments of a semiconductor device structure and a method for forming the same are provided. By forming back-side source/drain contact structure by a two-step etching back process, the defects formed in the bottom-up deposition process may be prevented, and the resistance and the yield may be improved.
(118) It should be noted that the gate-all-around (GAA) structure shown in the present disclosure is merely an example, the present disclosure may also be applied to FinFET devices or nanowire devices.
(119) In some embodiments, a method for forming a semiconductor device structure is provided. The method for forming a semiconductor device structure includes forming nanostructures over the front side of the substrate. The method for forming a semiconductor device structure also includes forming a gate structure surrounding the nanostructures. The method for forming a semiconductor device structure also includes forming a source/drain structure beside the gate structure. The method for forming a semiconductor device structure also includes forming a trench though the substrate from the back side of the substrate. The method for forming a semiconductor device structure also includes forming a first silicide layer in contact with the source/drain structure. The method for forming a semiconductor device structure also includes forming a second silicide layer over the first silicide layer and sidewalls of the trench. The method for forming a semiconductor device structure also includes depositing a first conductive material over the second silicide layer. The method for forming a semiconductor device structure also includes etching back the first conductive material. The method for forming a semiconductor device structure also includes etching back the second silicide layer. The method for forming a semiconductor device structure

also includes depositing a second conductive material in the trench.
(120) In some embodiments, a method for forming a semiconductor device structure is provided. The method for forming a semiconductor device structure includes forming a fin structure over a first side of a substrate. The method for forming a semiconductor device structure also includes forming a gate structure over the fin structure. The method for forming a semiconductor device structure also includes forming source/drain structures over opposite sides of the gate structure. The method for forming a semiconductor device structure also includes flipping over the substrate. The method for forming a semiconductor device structure also includes forming a hard mask layer over a second side of the substrate. The method for forming a semiconductor device structure also includes forming a trench exposing the source/drain structures from the second side of the substrate. The method for forming a semiconductor device structure also includes forming a liner layer over sidewalls of the trench. The method for forming a semiconductor device structure also includes forming a first silicide layer at a bottom of the trench. The method for forming a semiconductor device structure also includes depositing a second silicide layer in the trench and over the hard mask layer. The method for forming a semiconductor device structure also includes depositing a first conductive layer over the second silicide layer. The method for forming a semiconductor device structure also includes removing the first conductive layer over the hard mask layer and upper sidewalls of the trench. The method for forming a semiconductor device structure also includes removing the second silicide layer to expose the top surface of the hard mask layer and the upper sidewalls of the liner layer. The method for forming a semiconductor device structure also includes depositing a second conductive layer in the trench. The method for forming a semiconductor device structure also includes forming an overburden layer over the second side of the substrate. The method for forming a semiconductor device structure also includes planarizing the second side of the substrate.
(121) In some embodiments, a semiconductor device structure is provided. The semiconductor device structure includes nanostructures formed over a first side of a substrate. The semiconductor device structure also includes a gate structure surrounding the nanostructures. The semiconductor device structure also includes source/drain epitaxial structures formed over opposite sides of the gate structure. The semiconductor device structure also includes a first silicide layer formed at a second side of the source/drain epitaxial structures. The semiconductor device structure also includes a source/drain contact structure comprising a first portion and a second portion formed over the second side of the source/drain epitaxial structures. The first portion of the source/drain contact structure comprises a second silicide layer surrounding a first conductive material, and the second portion of the source/drain contact structure includes a second conductive material.
(122) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method for forming a semiconductor device structure, comprising: forming nanostructures over a front side of a substrate; forming a gate structure surrounding the nanostructures; forming a source/drain structure beside the gate structure; forming a trench though the substrate from a back side of the substrate; forming a first silicide layer in contact with the source/drain structure; forming a second silicide layer over the first silicide layer and sidewalls of the trench; depositing a

first conductive material over the second silicide layer; etching back the first conductive material; etching back the second silicide layer; and depositing a second conductive material in the trench.
2. The method for forming the semiconductor device structure as claimed in claim 1, further comprising: forming a liner layer in the trench before forming the first silicide layer; and removing the liner layer over a bottom surface of the trench.
3. The method for forming the semiconductor device structure as claimed in claim 1, further comprising: forming a hard mask layer over the back side of the substrate before forming the trench; depositing an overburden layer over the second conductive material and the hard mask layer; and planarizing the back side of the substrate to expose the hard mask layer.
4. The method for forming the semiconductor device structure as claimed in claim 1, wherein the first silicide layer and the second silicide layer comprise different materials.
5. The method for forming the semiconductor device structure as claimed in claim 1, further comprising: implanting a dopant at the back side of the substrate after depositing the second conductive material.
6. The method for forming the semiconductor device structure as claimed in claim 5, wherein the dopant comprises Ge, F, C, or a combination thereof.
7. The method for forming the semiconductor device structure as claimed in claim 1, further comprising: forming a barrier layer over the second silicide layer before depositing a first conductive material; and etching back the barrier layer when etching back the first conductive material.
8. A method for forming a semiconductor device structure, comprising: forming a fin structure over a first side of a substrate; forming a gate structure over the fin structure; forming source/drain structures over opposite sides of the gate structure; flipping over the substrate; forming a hard mask layer over a second side of the substrate; forming a trench exposing the source/drain structures from the second side of the substrate; forming a liner layer over sidewalls of the trench; forming a first silicide layer at a bottom of the trench; depositing a second silicide layer in the trench and over the hard mask layer; depositing a first conductive layer over the second silicide layer; removing the first conductive layer over the hard mask layer and upper sidewalls of the trench; removing the second silicide layer to expose a top surface of the hard mask layer and upper sidewalls of the liner layer; depositing a second conductive layer in the trench; forming an overburden layer over the second side of the substrate; and planarizing the second side of the substrate.
9. The method for forming the semiconductor device structure as claimed in claim 8, wherein the second conductive layer is deposited using a bottom-up deposition process.
10. The method for forming the semiconductor device structure as claimed in claim 8, wherein the first conductive layer and the second silicide layer are removed using different chemicals.
11. The method for forming the semiconductor device structure as claimed in claim 8, wherein the second conductive layer protrudes over the hard mask layer after depositing the second conductive layer.
12. The method for forming the semiconductor device structure as claimed in claim 8, wherein the second conductive layer has a curved upper surface before planarizing the second side of the substrate.
13. The method for forming the semiconductor device structure as claimed in claim 8, wherein the first conductive layer at the bottom of the trench is thicker than the first conductive layer over the hard mask layer after depositing a first conductive layer.
14. A method for forming a semiconductor device structure, comprising: forming a source/drain epitaxial structure over a front side of a substrate; forming a source/drain contact structure over the source/drain epitaxial structure; forming a trench though the substrate from a back side of the substrate to expose the source/drain epitaxial structure; forming a first silicide layer at a bottom of the trench; forming a second silicide layer over the first silicide layer; forming a first conductive layer surrounded by the second silicide layer; and forming a second conductive layer on the first

conductive layer and the second silicide layer.
15. The method for forming the semiconductor device structure as claimed in claim 14, further comprising: forming a third silicide layer between the source/drain contact structure and the source/drain epitaxial structure.
16. The method for forming the semiconductor device structure as claimed in claim 14, wherein a top surface of the first conductive layer is higher than a top surface of the second silicide layer.
17. The method for forming the semiconductor device structure as claimed in claim 14, wherein the first conductive layer has a concave top surface.
18. The method for forming the semiconductor device structure as claimed in claim 14, wherein the first conductive layer has a flat top surface.
19. The method for forming the semiconductor device structure as claimed in claim 14, wherein the second silicide layer has a flat top surface.
20. The method for forming the semiconductor device structure as claimed in claim 14, wherein the first conductive layer is separated from the first silicide layer by the second silicide layer.